

AKHILESH KUMAR MISHRA

SOFTWARE ENGINEER / FULL STACK ENGINEER

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EXPERIENCE

Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Re-architected monolithic backend services into distributed microservices using **Spring Boot**, **FastAPI**, and **Node.js**, improving performance for **2 million+ users**.
- Architected transactional endpoints for AEPS, MATM, and UPI platforms, managing **500,000+ daily transactions** with an **~80% success rate**.
- Secured backend APIs utilizing **JWT Authentication** and **RBAC**, integrating device binding and OTP security, reducing unauthorized access attempts by 99.9%.
- Architected and optimized databases using **MySQL**, **PostgreSQL**, and **MongoDB**, utilizing sharding and query tuning to reduce read/write latency by 45%.
- Established event-driven communication using **Kafka**, **Celery**, and **Redis** cache, boosting message processing throughput by 35%.
- Integrated transaction monitoring dashboards using **React.js**, **Redux**, and **WebSockets**, enabling real-time status updates under 500ms.

Freelance Software Engineer

Workforce Telemetry & Operations

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Created and developed a real-time workforce telemetry platform using React and FastAPI backend, serving **1,000+ active field agents**.
- Built an interactive live map dashboard in **React** using Ant Design to display speeds and alerts, reducing UI load times by 30%.
- Developed an asynchronous location ingestion pipeline utilizing **FastAPI**, **Celery**, and **Redis**, processing **10,000+ concurrent GPS pings** with **99.9% uptime**.

SELECTED PROJECTS

MERN FinTech & E-commerce Platform

A collection of full-stack web applications featuring secure user authentication, interactive dashboards, and REST APIs built with React, Node, Express, and MongoDB.

STACK: React.js / Node.js / Express.js / MongoDB / JWT / Redux

Facial Recognition Attendance System

Developed an automated student attendance tracking application integrating Python face detection algorithms (OpenCV and LBPH) with MySQL database.

STACK: Python / OpenCV / Tkinter / MySQL / Pandas

Himalayan Edges Website Development

Launched an e-commerce platform using React.js and Django with PWA capabilities, hosted on AWS, increasing page speed by 30%.

STACK: React.js / Django / SQLite / AWS / PWA

Traveling Portal & Landing Page

A fully responsive, modern front-end landing page for a travel agency built with HTML, CSS, and Bootstrap layouts.

STACK: HTML5 / CSS3 / JavaScript / Bootstrap

Persistent Task Management Dashboard

Programmed a desktop utility for task planning, category filtering, and JSON data persistence, featuring clean MVC architecture patterns.

STACK: Python / Tkinter / JSON / MVC Patterns

Real-Time Telemetry Geofencing Engine

Built a real-time workforce geolocation tracking engine with asynchronous GPS coordinate logs, map visualizations, and alerts via WebSockets.

STACK: React.js / FastAPI / Celery / Redis / PostgreSQL

SUMMARY

Versatile Software Development Engineer with extensive experience building secure, scalable full-stack web applications and microservice architectures across multiple technology stacks (MERN, Java Full Stack, Python Full Stack). Expert in system design, API development, and database engineering. Proven success in re-architecting monolithic services into containerized microservices, building real-time telemetry pipelines via WebSockets, and optimizing transaction-heavy ledger platforms. Led development teams and streamlined DevOps workflows using Docker and CI/CD.

EDUCATION

Bachelor of Technology (B.Tech.)

Kanpur Institute of Technology

2019 – 2023 | Kanpur, UP

SKILLS

LANGUAGES

Python | Java | JavaScript (ES6+) | TypeScript | SQL
C++ | HTML5 | CSS3

FRAMEWORKS

Spring Boot | Django | FastAPI | Node.js | Express.js
React.js | Next.js | Redux

DATA & MESSAGING

MySQL | PostgreSQL | MongoDB | Redis | Apache Kafka
Mongoose | Hibernate | JPA

DEVOPS & TOOLS

Docker | Git | GitHub Actions | CI/CD | Postman
Linux | OpenCV | Pandas | Maven

CERTIFICATIONS

Advanced MERN Stack Developer Certification
Database Systems & SQL Certification
Data Structures & Algorithms Certification
IIT Kanpur Cyber Security Certification

ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.